2025 AMC 10B Problem 13 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10B Problem 13. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10B Problem 13

Problem:

The altitude to the hypotenuse of a $30^\circ - 60^\circ - 90^\circ$ right triangle is divided into two segments of lengths $x < y$ by the median to the shortest side of the triangle. What is the ratio $\frac{x}{x+y}$?

Answer Choices:

A. $\frac{3}{7}$

B. $\frac{\sqrt{3}}{4}$

C. $\frac{4}{9}$

D. $\frac{5}{11}$

E. $\frac{4\sqrt{3}}{15}$

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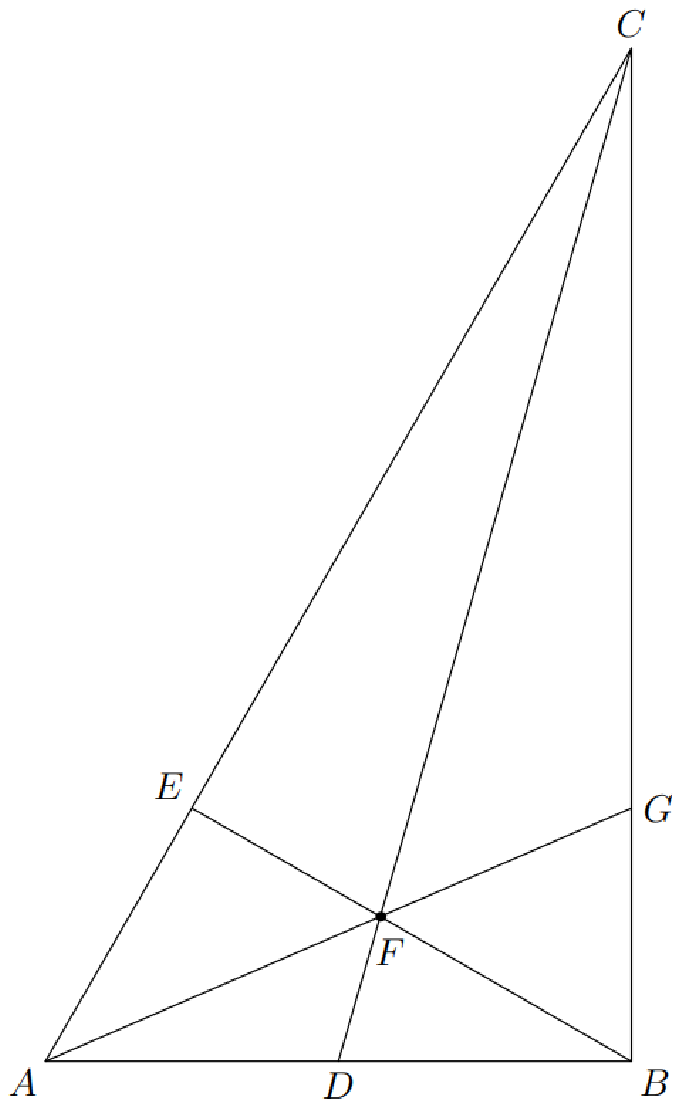
Solution 1:

We will use mass points.



Note that since $\triangle ABC \sim \triangle BEC$, we have

$$CE = \frac{BC^2}{CA} = \frac{3}{4}$$

as $\triangle ABC$ is a $30^\circ - 60^\circ - 90^\circ$ triangle.

Assign a mass of 3 at A, which means we will assign a mass of 1 at C. As $AD = DB$, this means B has a mass of 3 as well, so E has a mass of 4 and F has a mass of 7.

Thus

$$\frac{FB}{EF} = \frac{\text{mass at } E}{\text{mass at } B} = \frac{4}{3}$$

Thus $x = 3, y = 4$, and $\dfrac{x}{x+y} = \boxed{\textbf{(A) } \frac{3}{7}}$.

Solution 2:

While there is a simple solution with mass points, we defer to using areas.

Note that

$$\frac{[CFB]}{[CEF]} = \frac{\frac{1}{2} \cdot FB \cdot \text{distance from } C \text{ to } FB}{\frac{1}{2} \cdot EB \cdot \text{distance from } C \text{ to } EF} = \frac{FB}{EF}$$

as FB and EF are the same line. Note that

$$\frac{[CEF]}{[CAF]} = \frac{\frac{1}{2} \cdot FE \cdot CE}{\frac{1}{2} \cdot FE \cdot CA} = \frac{CE}{CA}$$

as FE is perpendicular to AC. Note that

$$[CAD] = \frac{1}{2} \cdot AD \cdot CB = \frac{1}{2} \cdot BD \cdot CB = [CBD]$$

$$[FAD] = \frac{1}{2} \cdot AD \cdot \text{distance from } F \text{ to } AD = \frac{1}{2} \cdot BD \cdot \text{distance from } F \text{ to } BD = [FBD]$$

so $[AFC] = [ACD] - [AFD] = [BCD] - [BFC] = [CFB]$. Thus, combining these we get

$$\frac{FB}{EF} = \frac{[CFB]}{[CEF]} = \frac{[AFC]}{[CEF]} = \frac{CA}{CE}$$

Note that since $\triangle ABC \sim \triangle BEC$, we have

$$CE = \frac{BC^2}{CA} = \frac{3}{4}$$

as $\triangle ABC$ is a $30 - 60 - 90$ triangle, hence

$$\frac{FB}{EF} = \frac{4}{3}$$

Thus $x = 3, y = 4$, and $\frac{x}{x+y} = \boxed{\textbf{(A) } \frac{3}{7}}$.

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